PhD Bibliography : Optimization of learning-based methods without ground truth and applicability to positron emision tomography

Sacha Bouchez-Delotte

March 11, 2026

Contents

1 Positron Emission Tomography (PET) 4

1.1 How PET works . . . 4

1.2 Deterministic reconstruction methods . . . 4

1.3 Iterative reconstruction in PET . . . 6

1.3.1 Poisson bayesian inference . . . 6

1.3.2 Maximum-Likelihood Expectation-Maximization (MLEM) . . . 6

1.3.3 Ordered Subset Expectation Maximization (OSEM) . . . 7

1.3.4 Regularized reconstruction in PET . . . 7

1.4 Deep learning for PET image reconstruction . . . 10

1.4.1 Supervised methods . . . 10

1.4.2 Unrolling techniques . . . 10

2 Denoising methods 11

2.1 Self-supervised denoising . . . 11

2.1.1 Denoising Auto-Encoders . . . 11

2.1.2 Score-based denoising . . . 12

2.2 Noise2Noise . . . 16

2.2.1 MSE loss case . . . 17

2.2.2 Poisson negative log-likelihood case . . . 18

2.2.3 Consensus loss . . . 20

3 Linear inverse problems 22

3.1 Generalization of Noise2Noise to linear inverse problems . . . 22

3.1.1 Poisson linear inverse problems . . . 24

Abstract

This doctoral project aims at proposing new statistical learning based methods to solve linear inverse problems, and more specifically image reconstruction. The focus will be laid on the possibility to perform the training stage without ground truth images, provided that suited sets of noisy data are available, as pioneered by the authors of Noise2noise. A second step of the project will be to consider Positron Emission Tomography (PET) imaging as a specific application case, in collaboration with Nantes CHU. In this context, list mode data can be split into statistically independent subsets in many ways, which could yield a tailored version of Noise2noise with enhanced properties.

1 Positron Emission Tomography (PET)

1.1 How PET works

1.2 Deterministic reconstruction methods

Radon transform

Definition 1.1. *Radon Transform The Radon transform is an integral transform that takes a two-dimensional function $f(x, y)$ and computes its line integrals along straight lines parameterized by angle ϕ and distance s from the origin. Mathematically, the Radon transform is defined as:*

$$g(s, \phi) = \int \int_{-\infty}^{\infty} f(x, y)\, \delta(s - x\cos\phi - y\sin\phi)\, dxdy \tag{1}$$

In the context of PET imaging, the Radon transform models the process of acquiring projection data (i.e., sinograms) from the distribution of radiotracer within the body.

Theorem 1.1. *Central Slice Theorem The Radon transform is invertible in the frequency domain.*

Proof. We consider the 1D Fourier transform of the projection $g(s, \phi)$ with respect to the variable s:

$$G(\omega, \phi) = \int_{-\infty}^{\infty} g(s, \phi) e^{-i\omega s} ds$$

Substituting the expression for $g(s, \phi)$ into the Fourier transform, we get:

$$G(\omega, \phi) = \int_{-\infty}^{\infty} \left(\int \int_{-\infty}^{\infty} f(x, y)\, \delta(s - x\cos\phi - y\sin\phi)\, dxdy \right) e^{-i\omega s} ds$$

Interchanging the order of integration, we have:

$$G(\omega, \phi) = \int \int_{-\infty}^{\infty} f(x, y) \left(\int_{-\infty}^{\infty} \delta(s - x\cos\phi - y\sin\phi) e^{-i\omega s} ds \right) dxdy$$

The inner integral evaluates to $e^{-i\omega(x\cos\phi + y\sin\phi)}$, leading to:

$$G(\omega, \phi) = \int \int_{-\infty}^{\infty} f(x, y) e^{-i\omega(x\cos\phi + y\sin\phi)} dxdy$$

This expression represents the 2D Fourier transform of $f(x, y)$ evaluated at the frequency coordinates $(\omega\cos\phi, \omega\sin\phi)$. Thus, we have:

$$G(\omega, \phi) = F(\omega\cos\phi, \omega\sin\phi) \tag{2}$$

where $F(u, v)$ is the 2D Fourier transform of $f(x, y)$. This shows that the 1D Fourier transform of the projection at angle ϕ corresponds to a slice of the 2D Fourier transform of the object $f(x, y)$ along the direction defined by ϕ. $\square$

This theorem comes with limitations in practice as we can only acquire a limited number of projections and the data is noisy. Indeed, going from the frequency domain to the spatial domain requires interpolation, which can introduce artifacts if the sampling is not dense enough. Hence, it is not possible to perfectly reconstruct the original image using only the Radon transform and its inverse.

Filtered Back Projection (FBP) Another possible reconstruction method is known as Back Projection (BP). This method consists in smearing back each projection across the image along the direction it was acquired.

Definition 1.2. *Back Projection The Back projection operation b is defined as follows:*

$$b(x, y) = \int_0^\pi g(s, \phi) d\phi = \int_0^\pi g(x \cos\phi + y \sin\phi, \phi) d\phi \tag{3}$$

Proposition 1.1. *The Back Projection operation can be interpreted as a convolution of the original image with a $1/r$ kernel, where r is the distance from the center of the image. This convolution results in a blurring effect, as high-frequency components of the image are attenuated*

Proof. Taking the 2D Fourier transform of the Back Projection operation, we have:

$$\begin{aligned}
B(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_0^\pi g(s, \phi) e^{-2i\pi(ux+vy)} dx dy d\phi \\
&= \int_0^\pi \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(s, \phi) e^{-2i\pi(ux+vy)} dx dy \right) d\phi \\
&= \int_0^\pi \left(\int_{-\infty}^{\infty} g(s, \phi) e^{-2i\pi s\sqrt{u^2+v^2}} ds \right) d\phi \\
&= \int_0^\pi G(\sqrt{u^2 + v^2}, \phi) d\phi
\end{aligned}$$

Substituting the expression for $G(\omega, \phi)$ from the Central Slice Theorem (1.1), we get:

$$B(u, v) = \int_0^\pi F(\sqrt{u^2 + v^2} \cos\phi, \sqrt{u^2 + v^2} \sin\phi) d\phi$$

This integral represents a radial integration of the 2D Fourier transform of the original image $f(x, y)$. The result is that the Back Projection operation effectively applies a low-pass filter to the original image, attenuating high-frequency components and resulting in a blurred image. $\square$

To counteract this blurring effect, a filtering step is introduced before the Back Projection, leading to the Filtered Back Projection (FBP) method.

Definition 1.3. *Filtered Back Projection The Filtered Back Projection operation consists in applying a high-pass filter to the projections before performing the Back Projection:*

$$f_{FBP}(x, y) = \int_0^{\pi} (g(s, \phi) * h(s))\, d\phi \tag{4}$$

where $h(s)$ is a high-pass filter kernel, such as the Ram-Lak filter.

1.3 Iterative reconstruction in PET

1.3.1 Poisson bayesian inference

In PET imaging, the data acquisition process can be modeled as a Poisson process as the number of detected events in each detector pair follows a Poisson distribution. Thus, the likelihood of observing the measured data y given the image x can be expressed as:

$$p(y|x) = \prod_{i=1}^{d} \frac{(y_i(x))^{y_i} e^{-y_i(x)}}{y_i!} \tag{5}$$

where $y_i(x) = (Ax)_i$ is the expected number of counts in detector pair i given the image x. It is modeled as a linear operation of the image through the system matrix A.

We want to estimate the image x that maximizes the likelihood of the unobserved data x given the observed data y. This is known as the Maximum Likelihood (ML) estimation. We will transform this formulation into a minimization problem by taking the negative log-likelihood:

$$\hat{x}_{\mathrm{ML}} = \arg\min_x \sum_{i=1}^{d} ((Ax)_i - y_i \log(Ax)_i) \tag{6}$$

1.3.2 Maximum-Likelihood Expectation-Maximization (MLEM)

Among iterative algorithms for PET reconstrcution, the **Maximum-Likelihood Expectation-Maximization (MLEM)** proposed by [Shepp et al. 1982] is one of the most widely used. It is an iterative method to find the ML estimate of the image x given the measured data y.

The negative log-likelihood is convex, thus the MLEM algorithm is guaranteed to converge to a global minimum, providing an estimate of the image x that is consistent

with the measured data y under the Poisson noise model. However, it can be slow to converge and can produce noisy images if the number of iterations is too high. In order to address this, iterations can be stopped early, or output can be post-processed or post-filtered.

Algorithm 1 Maximum-Likelihood Expectation-Maximization (MLEM)

Input: Measured data y, system matrix A, initial estimate $x^{(0)}$
Output: Estimated image $x^{(k)}$
for $k = 0$ to $K - 1$ **do**
    E-step: Compute the expected number of counts

$$\hat{y}_i^{(k)} = (Ax^{(k)})_i$$

    M-step: Update the image estimate

$$x_j^{(k+1)} = x_j^{(k)} \frac{\sum_{i=1}^{d} A_{ij} \frac{y_i}{\hat{y}_i^{(k)}}}{\sum_{i=1}^{d} A_{ij}}$$

end for
Return $x^{(K)}$

1.3.3 Ordered Subset Expectation Maximization (OSEM)

In order to speed up the convergence of the MLEM algoritm, the **Ordered Subset Expectation Maximization (OSEM)** algorithm was proposed by [Hudson et al. 1994]. It is a variant of the MLEM algorithm that divides the measured sinogram y into subsets $\{y_1, \ldots, y_Q\}$ and performs the E-step and M-step from 1 on each subset sequentially. Algorithm is given in 2.

As the algorithm is not designed to minimized a well-defined cost funcion, the latter does not lead to the same ML estimate of 6 as the MLEM algorithm. However, it is much faster to converge and can be used in practice with a large number of subsets without significant loss of image quality. Post-processing or post-filtering can also be applied to the output of the OSEM algorithm to further improve image quality.

1.3.4 Regularized reconstruction in PET

This reconstruction problem can also be formulatted as a regularized optimization by introducing some prior knowledge on the image x. Indeed, Bayes's rule gives us

Algorithm 2 Ordered Subset Expectation Maximization (OSEM)

Input: Measured data y, system matrix A, initial estimate $x^{(0)}$, number of subsets Q
Output: Estimated image $x^{(k)}$
for $k = 0$ to $K - 1$ **do**
  for $q = 1$ to Q **do**
    E-step: Compute the expected number of counts for subset q

$$\hat{y}_i^{(k,q)} = (Ax^{(k)})_i \quad \text{for } i \in \text{subset } q$$

    M-step: Update the image estimate using subset q

$$x_j^{(k,q)} = x_j^{(k,q-1)} \frac{\sum_{i \in \text{subset } q} A_{ij} \frac{y_i}{\hat{y}_i^{(k,q)}}}{\sum_{i \in \text{subset } q} A_{ij}}$$

  end for
end for
Return $x^{(K)}$

the following expression for the posterior distribution of x given y:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} \tag{7}$$

where $p(x)$ is the prior distribution of the image x and $p(y)$ is the marginal likelihood of the data y. Once agin, by taking the negative log-likelihood of $p(x|y)$, we can transform the problem into a penalized likelihood minimization problem:

$$\hat{x}_{\text{MAP}} = \arg\min_x \left(-\log p(y|x) - \log p(x)\right) \tag{8}$$

where $-\log p(y|x)$ is the data fidelity term and $-\log p(x)$ is the regularization term that encodes our prior knowledge on the image x. This prior can be expressed in the following exponential form :

$$p(x) \propto e^{-\beta R(x)} \tag{9}$$

where $R(x)$ is a regularization function and β is a regularization parameter that controls the trade-off between data fidelity and regularization. The optimisation hence becomes:

$$\hat{x}_{\text{MAP}} = \arg\min_x \left(-\log p(y|x) + \beta R(x)\right) \tag{10}$$

The penalized likelihood approach allows us to incorporate various types of prior knowledge about the image, such as smoothness, sparsity, or anatomical information. The choice of the regularization function $R(x)$ must alleviate the ill-posedness of the reconstruction problem while preserving important features of the image, such as edges and textures. The following paragraph presents some commonly used regularization functions in PET image reconstruction.

Markov Random Fields **Markov Random Fields (MRFs)** are a class of probabilistic graphical models that can be used to model spatial dependencies in images. In the context of PET image reconstruction, MRFs can be used to define a prior distribution over the image x that encourages spatial smoothness while preserving edges. Its most basinc formulation can be expressed as follows:

$$R(x) = \sum_{j} \sum_{k \in N(j)} w_{jk} \phi(x_j, x_k) \tag{11}$$

where $N(j)$ is the set of neighboring pixels of pixel j, w_{jk} is a weight that controls the strength of the interaction between pixels j and k, and ϕ is a potential function that defines the type of regularization applied to the differences between neighboring pixels.

l_2 **penalty (Quadratic)** One of the most basic choice for the potential function ϕ is the quadratic function, first proposed by [Tikhonov et al. 1977], known as **Tikhonov regularization**. It encourages global smoothness by penalizing the squared differences between neighboring pixels. The side effect of this regularization is that it can oversmooth the image and blur edges, which are important features in medical imaging as we want to distinguish between different tissues and structures.

$$\phi(x_j, x_k) = (x_j - x_k)^2 \tag{12}$$

l_1 **penalty (Total Variation)** In order to preserve edges while still encouraging smoothness, the **Total Variation (TV)** regularization was proposed by [Rudin et al. 1992]. It penalizes the absolute differences between neighboring pixels, which allows for sharp transitions in the image while still reducing noise in homogeneous regions.

$$\phi(x_j, x_k) = |x_j - x_k| \tag{13}$$

Huber penalty Hybrid penalties combine the advantages of both l_2 and l_1 regularization by applying a quadratic penalty to small differences and a linear penalty to large differences. This allows for smoothness in homogeneous regions while preserving

edges. One example of a hybrid penalty is the Huber function, which is defined as follows:

$$\phi(x_j, x_k) = \begin{cases} \dfrac{(x_j - x_k)^2}{2\delta} & \text{if } |x_j - x_k| \leq \delta \\ |x_j - x_k| - \dfrac{\delta}{2} & \text{if } |x_j - x_k| > \delta \end{cases} \tag{14}$$

Gibbs penalty The Gibbs penalty is a concave prior that has been shown to be effective in PET imaging [Nuyts et al. 2002]. It is also known as the **Relative Differences**.

$$\phi(x_j, x_k) = \frac{(x_j - x_k)^2}{x_j + x_k + \gamma|x_j - x_k|} \tag{15}$$

where the γ parameter controls which relative diferences are considered as large.

1.4 Deep learning for PET image reconstruction

1.4.1 Supervised methods

1.4.2 Unrolling techniques

2 Denoising methods

In this section, we present several denoising methods for a certain signal x corrupted by noise to produce a noisy observation y. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ and $Y : \Omega \to \mathbb{R}^n$ be random variables representing respectively the clean signal and the noisy observation. We consider a corruption process $\mathcal{C}$ that maps the clean signal X to the noisy observation Y according to a certain noise model. This corruption process can be expressed as follows :

$$Y|X \sim \mathcal{C}(X) \tag{16}$$

Additive gaussian noise We will consider the residual noise B defined as follows :

$$B = Y - X \tag{17}$$

2.1 Self-supervised denoising

2.1.1 Denoising Auto-Encoders

Whereas traditional auto-encoders are trained to reconstruct their input through an encoder-decoder structure, Denoising auto-encoders (DAEs) are trained to reconstruct clean data from corrupted versions of it. They have been first introduced by [Vincent et al. 2008] and have since been widely used in various applications such as image denoising, inpainting or other inverse problems. The DAE objective is thus :

$$\mathcal{L}_{DAE}(\theta) = \mathop{\mathbb{E}}_{\substack{X \sim p_X \\ Y \sim N(X)}} \left[\|X - f_\theta(Y)\|_2^2 \right] \tag{18}$$

with N being a random corruption process.

Proposition 2.1. *A neural network $f = D_\sigma$ trained under the DAE objective estimates the following quantity :*

$$D_\sigma(y) \approx \mathbb{E}\left[X|Y = y\right] \tag{19}$$

Proof. We have :

$$\begin{aligned} D_\sigma(y) &\approx \arg\min_z \mathbb{E}_{X|Y=y} \left[\|X - z\|_2^2 \right] \\ &= \arg\min_z \int \|x - z\|_2^2 p(x|y) dx \end{aligned}$$

Taking the derivative with respect to z and setting it to zero gives us :

$$\frac{\partial}{\partial z}\int \|x-z\|_2^2 p(x|y)dx = \int 2(z-x)p(x|y)dx = 0$$
$$\Rightarrow z = \int xp(x|y)dx = \mathbb{E}[X|Y=y]$$

□

2.1.2 Score-based denoising

Definition 2.1. *Score function The score function of a random variable X with probability density function $p(x)$ is defined as the gradient of the log-density with respect to x :*

$$s_X(x) = \nabla_x \ln p(x) \tag{20}$$

This function denotes the direction of maximum increase of the probability density function at point x.

Score matching estimation As first introduced by [Hyvärinen 2005], score matching is a method to estimate parametric models of probability density functions without requiring the computation of the normalizing constant. Given a parametric model score function $s(\cdot;\theta)$ with parameter θ, and the data score function $s_X(\cdot)$ of observed data sampled from X, we want to minimize the distance between the data score function $s_X(\cdot)$ and the model score function $s_X(\cdot;\theta)$ using the following objective :

$$\mathcal{L}_{SM}(\theta) = \mathbb{E}_X\left[\|s_X(x) - s(x;\theta)\|_2^2\right] \tag{21}$$

Minimization of this objective does not require knowledge of the normalizing constant of $s(\cdot;\theta)$ as it only involves derivatives of the log-density. They also show that we don't need to know the data score function $s_X(x)$ to optimize this objective as it can be rewritten as :

$$\mathcal{L}_{SM}(\theta) = \mathbb{E}_X\left[Tr\left(\nabla_x s(x;\theta)\right) + \frac{1}{2}\|s_X(x;\theta)\|_2^2\right] + C \tag{22}$$

This is very convenient as it avoids the need to estimate the data score function over the observed data, which is generally intractable.

Denoising Auto-Encoders and score function estimation DAEs and score matching principles have been linked together by [Vincent 2011]. They built an objective similar to score matching but defined on clean/noisy image pair with the corruption process N distribution being known.

$$\mathcal{L}_{DSM}(\theta) = \mathop{\mathbb{E}}_{X\sim p_X Y\sim N(X)} \left[\|s_X(Y,\theta) - s_{Y|X}(Y|X)\|_2^2\right] \tag{23}$$

They prove the specific case of Tweedie's formula for additive gaussian noise (see Theorem 2.1) which will be considered in the following.

[Alain et al. 2014] further prove that when the gaussian noise level σ tends to zero, the DAE learns the score of the data distribution as follows :

$$f_{\theta^*}(y) = y + \sigma^2 s_X(y) + \underset{\sigma\to 0}{o}(\sigma^2) \tag{24}$$

This can be generalized to a wider range of exponential family distributions thanks to Tweedie's formula [Efron 2011]. We retain the two following theorems that stand for additive gaussian noise and Poisson noise respectively.

Theorem 2.1. *Tweedie's formula for additive gaussian noise Let X be a random variable and $Y = X+N$ be a noisy observation of X corrupted by an additive gaussian noise N independent from X with variance σ^2. The MMSE estimator of X given $Y = y$ is given by :*

$$\mathbb{E}\left[X|Y = y\right] = y + \sigma^2 \nabla_y \ln p(y) \tag{25}$$

Proof. Bayes' rule gives us:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

The likelihood $p(y|x)$ is given by the noise distribution:

$$p(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{||y-x||^2}{2\sigma^2}\right)$$

Thus we have:

$$p(x|y) = \frac{1}{Z(y)} \exp\left(-\frac{||y-x||^2}{2\sigma^2}\right) p(x)$$

with $Z(y)$ acting as a normalizing constant:

$$Z(y) = \int p(x) \exp\left(-\frac{||y-x||^2}{2\sigma^2}\right) dx$$

Then we compute the derivative of $p(y)$:

$$\begin{aligned}
\nabla_y p(y) &= \frac{d}{dy}\int p(y|x)p(x)dx \\
&= \int \frac{\partial}{\partial y} p(x)\exp\left(-\frac{||y-x||^2}{2\sigma^2}\right)dx \\
&= -\frac{1}{\sigma^2}\int (y-x)p(x)\phi_\sigma(y-x)dx
\end{aligned}$$

where we denote ϕ_σ the gaussian kernel of standard deviation σ.

$$\begin{aligned}
\nabla_y \ln p(y) = \frac{p'(y)}{p(y)} &= -\frac{1}{\sigma^2}\left(\int (y-x)\frac{p(x)\phi_\sigma(y-x)}{p(y)}dx\right) \\
&= -\frac{1}{\sigma^2}\left(y - \int xp(x|y)dx\right) \\
&= -\frac{1}{\sigma^2}(y - \mathbb{E}[x|y])
\end{aligned}$$

□

Theorem 2.2. *Tweedie's formula for Poisson noise Let X be a random variable and Y be a noisy observation of X corrupted by poissonian noise. The MMSE estimator of X given $Y = y$ is given by :*

$$\mathbb{E}\left[X|Y=y\right] = (y+1)\exp\left(\nabla_y \ln p(y)\right) \tag{26}$$

Proof. The assumption of Poisson noise means that the likelihood $p(y|x)$ is given by:

$$p(y|x) = \frac{(x)^y e^{-x}}{y!}$$

Then we compute the derivative of $p(y)$:

$$\begin{aligned}
\exp(\ln \nabla_y p(y)) &= \frac{p(y+1)}{p(y)} \\
&= \frac{1}{p(y)}\int p(x)\frac{x}{y+1}\frac{x^y}{y!}e^{-x}dx \\
&= \frac{1}{y+1}\int x\frac{p(x)p(y|x)}{p(y)}dx \\
&= \frac{1}{y+1}\mathbb{E}[x|y]
\end{aligned}$$

Thus we have:

$$\mathbb{E}[x|y] = (y+1)\exp(\ln \nabla_y p(y))$$

□

Given the previous proposition and theorems, one can build a link between denoising auto-encoders and score function estimation as follows :

$$s_Y(y) = \frac{D_\sigma(y) - y}{\sigma^2} \quad \text{(Gaussian noise)} \tag{27}$$

$$s_Y(y) = \ln\left(\frac{D_{Poisson}(y)}{y+1}\right) \quad \text{(Poisson noise)} \tag{28}$$

We can highlight an intriguing application of [Kim et al.] who leverage this link to estimate the score function using a DAE trained under an additive gaussian noise assumption, even when the actual noise model is not gaussian. They then use the estimated score function to denoise images corrupted by other types of noise using the appropriate variant of Tweedie's formula.

Noise2Void / Noise2Self The Noise2Void [Krull et al. 2019] and Noise2Self [Batson et al.] frameworks further extend the idea of training denoising models without clean targets by using only single noisy images. [Batson et al.] introduce the concept of $\mathcal{J}$*-invariant* functions, which are functions that do not depend on certain subsets of their input. We train a denoising model to be $\mathcal{J}$-invariant by masking certain pixels of the input image and predicting their values based on the surrounding pixels. The work of [Krull et al. 2019] implements this idea by randomly masking pixels in the input image during training and using the unmasked pixels to predict the masked ones. This is called the *blind-spot* strategy and ensures that the model does not learn to simply copy the noisy input to the output, thus learning the identity.

Noisier2Noise In order to leverage the lack of paired noisy images, [Moran et al. 2019] propose the Noisier2Noise framework which consists in adding extra noise to a single noisy image to create a pair of noisy images. The model is then trained to map the more noisy image to the less noisy one, learning the underlying noise model in the process. This learned noise model is then assumed to retrieve the clean image from a noisy observation at inference time.

Noise2Score The authors in [Kim et al.] propose a 2-step framework called Noise2Score. First the score function is being estimated using a DAE trained under framework closed to Noisier2Noise [Moran et al. 2019] where noise is embedded into the images according to the noise model and then the network learn the mapping to retrieve less

noisy images. Then, the estimated denoised images are simply computed using the appropriate Tweedie's formula variant. This method is very promissing at it is at the crossroad between bayesian methods with strong mathematical background and self-supervised learning.

2.2 Noise2Noise

A seminal work in the denoising field is the Noise2Noise framework [Lehtinen et al. 2018]. Noise2Noise is a denoising method at the crossroad between self-supervised methods — which learn to denoise only from noisy samples — and supervised methods — which require clean targets to learn the mapping between noisy and clean image. The key idea of Noise2Noise is that we learn to denoise images from pairs of noisy images instead of noisy-clean image pairs or single noisy images. These pairs of noisy images can be obtained in various ways depending on the application. Each pair must represent the same underlying clean image but with different realizations of the noise process. Hence, we use the information contained in the noisy data itself to learn how to denoise it, without ever observing clean images.

This section will use the formalism of 16 to present the Noise2Noise framework and its theoretical guarantees. Let $Y^{(1)}$ and $Y^{(2)}$ be two noisy observations of the same underlying clean image X obtained through the same corruption process $\mathcal{C}$ but with different realizations of the noise. Thus, the noisy pairs are generated according to the following noise model :

$$\begin{cases} Y^{(j)}|X \sim \mathcal{C}(X), & j = 1, 2 \\ Y^{(j)} = X + B^{(j)}, & j = 1, 2 \end{cases} \tag{29}$$

Let $\{(Y_i^{(1)}, Y_i^{(2)})\}_{i=1}^N$ be a finite sequence of independant and statistically distributed random variables. For $\omega_i \in \Omega$, we denote $Y_i^{(j)}(\omega_i) = y_i^{(j)} = x_i + b_i^{(j)}$ the observed noisy images for $j = 1, 2$ and $i = 1, \ldots, N$.

We train a neural network f_θ with parameters θ to map the noisy image $y_i^{(1)} = x_i + b_i^{(1)}$ to the other noisy image $y_i^{(2)} = x_i + b_i^{(2)}$ both representing a different corruption of the same underlying clean image x_i and vice versa by minimizing the following loss function :

$$\mathcal{L}_{\text{N2N}}(\theta) = \frac{1}{2N} \sum_{i=1}^{N} \left[\rho(f_\theta(y_i^{(1)}), y_i^{(2)}) + \rho(f_\theta(y_i^{(2)}), y_i^{(1)}) \right] \tag{30}$$

with ρ a loss function such as the Mean Squared Error (MSE), the Mean Absolute Error (MAE) or any other appropriate loss.

The supervised counterparts of 30 would be :

$$\mathcal{L}_{\text{clean}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \rho(f_\theta(y_i), x_i) \tag{31}$$

with $y_i = Y(\omega_i)$ being a single noisy observation of the clean image x_i.

2.2.1 MSE loss case

Theorem 2.3. *Minimizing 30 with $\rho(x, y) = \|x - y\|_2^2$ gives the same estimator as minimizing 31 as $N \to \infty$ under the following conditions :*

- *$Y^{(1)}$ and $Y^{(2)}$ are independent given the clean image X (i.e. $Y^{(1)}|X \perp\!\!\!\perp Y^{(2)}|X$).*
- *The noise has zero mean, ie $\mathbb{E}[B^{(j)}] = 0$ for $j = 1, 2$.*

Proof. We expand the Noise2Noise empirical risk as follows :

$$\begin{aligned}
R_{N,1}^{n2n}(\theta) &= \frac{1}{N}\sum_{i=1}^{N}\left[\|f_\theta(Y_i^{(1)}) - (X^{(i)} + B^{(2)})\|_2^2\right] \\
&= \frac{1}{N}\sum_{i=1}^{N}\left[\|(f_\theta(Y_i^{(1)}) - X^{(i)}) - B^{(2)}\|_2^2\right] \\
&= R_{N,1}^{clean}(\theta) + \frac{1}{N}\sum_{i=1}^{N}\left[\|B^{(2)}\|_2^2\right] - \frac{2}{N}\sum_{i=1}^{N}\left[(f_\theta(Y_i^{(1)}) - X^{(i)})^T B^{(2)}\right]
\end{aligned}$$

However, $\frac{2}{N}\sum_{i=1}^{N}\left[(f_\theta(Y_i^{(1)}) - X^{(i)})^T B^{(2)}\right] \xrightarrow[N\to\infty]{a.s.} 2\mathbb{E}\left[(f_\theta(Y^{(1)}) - X)^T B^{(2)}\right]$ due to the law of large numbers. This expectation vanishes under the independence and zero-mean assumptions on the noise. Thus, we have :

$$R_{N,1}^{n2n}(\theta) = R_{N,1}^{clean}(\theta) + \frac{1}{N}\sum_{i=1}^{N}\left[\|B^{(2)}\|_2^2\right] + o(1)$$

We show a similar result for the second term of the Noise2Noise empirical risk by swapping $Y^{(1)}$ and $Y^{(2)}$ and we get the following result :

$$R_N^{n2n}(\theta) = 2R_N^{clean}(\theta) + \frac{1}{N}\sum_{i=1}^{N}\left[\|B_i^{(1)}\|_2^2 + \|B_i^{(2)}\|_2^2\right] + o(1)$$

where we are left with the standard clean denoising empirical risk with its underlying bias and variance decomposition, and an irreductible noise variance term which has no influence as it is independent of θ. □

Remark 2.1. *The previous proposition can be extended to other loss functions such as the MAE loss $\rho = \|\cdot\|_1$ under the assumption that the noise distribution is symmetric with respect to zero (i.e. $B^{(j)}$ and $-B^{(j)}$ have the same distribution for $j = 1, 2$). In this case, the network learns to estimate the median of the underlying clean image distribution instead of the mean as in the MSE case.*

2.2.2 Poisson negative log-likelihood case

Definition 2.2. *For two random variables $X : \Omega \to \mathbb{R}^d$ and $Y : \Omega \to \mathbb{R}^d$, we say that $Y \mid X \sim$ Poisson(X) if :*

$$\forall i = 1 \dots d, \quad \mathbb{P}(Y[i] = k \mid X[i] = x[i]) = \frac{(x[i])^k e^{-x[i]}}{k!}, \quad k \in \mathbb{N}$$

Proposition 2.2. *Under the assumption $Y \mid X \sim$ Poisson(X), we have :*

$$\mathbb{E}[Y \mid X] = X$$

Proof. By definition of the Poisson distribution, we have :

$$\begin{aligned}
\mathbb{E}[Y[i] \mid X[i] = x[i]] &= \sum_{k=0}^{\infty} k \cdot \mathbb{P}(Y[i] = k \mid X[i] = x[i]) \\
&= \sum_{k=0}^{\infty} k \cdot \frac{(x[i])^k e^{-x[i]}}{k!} \\
&= x[i] \sum_{k=1}^{\infty} \frac{(x[i])^{k-1} e^{-x[i]}}{(k-1)!} \\
&= x[i] \cdot 1 = x[i]
\end{aligned}$$

Thus, we have $\mathbb{E}[Y \mid X] = X$. □

Now, we consider the case of Poisson noise (i.e. $Y^{(j)} \mid X^{(j)} \sim \text{Poisson}(X^{(j)}), \quad j = 1, 2$). Given a clean image x and a noisy observation y corrupted by Poisson noise, the likelihood of observing y given x is given by :

$$p(y|x) = \prod_i \frac{(x[i])^{y[i]} e^{-x[i]}}{y[i]!}$$

Taking the negative logarithm of this likelihood gives us the Poisson negative log-likelihood loss as defined in the following definition.

Definition 2.3. *The Poisson negative log-likelihood loss between two images y and x is defined as follows :*

$$\mathcal{L}_{Poisson}(y, x) = \sum_i [y[i] - x[i]\log(y[i]) + \log(x[i]!)]$$

Theorem 2.4. *If $Y|X = x \sim Poisson(x)$, then minimizing 30 with $\rho = \mathcal{L}_{Poisson}$ gives the same estimator as minimizing 31 as $N \to \infty$ under the following conditions :*

- *$Y^{(1)}$ and $Y^{(2)}$ are independent given the clean image X (i.e. $Y^{(1)}|X \perp\!\!\!\perp Y^{(2)}|X$)..*

Proof. The poisson negative log-likelihood between Y_1 and Y_2 can be expressed as follows :

$$\mathcal{L}_{Poisson}(Y_1, Y_2) = Y_1 - Y_2\log(Y_1) + \log(Y_2!)$$

We expand the Noise2Noise Poisson empirical risk as follows :

$$\begin{aligned} R^{n2n}_{N,1}(\theta) &= \frac{1}{N}\sum_{i=1}^{N}\left[\mathcal{L}_{Poisson}(f_\theta(Y_i^{(1)}), Y_i^{(2)})\right] \\ &= \frac{1}{N}\sum_{i=1}^{N}\left[f_\theta(Y_i^{(1)}) - Y_i^{(2)}\log(f_\theta(Y_i^{(1)})) + \log(Y_i^{(2)}!)\right] \end{aligned}$$

Yet, $\frac{1}{N}\sum_{i=1}^{N} Y_i^{(2)}\log(f_\theta(Y_i^{(1)})) \xrightarrow[N\to\infty]{a.s.} \mathbb{E}\left[Y^{(2)}\log(f_\theta(Y^{(1)}))\right]$ due to the law of large numbers. Under the independence assumption on the noise, this expectation can be rewritten as

$$\mathbb{E}\left[Y^{(2)}|X\right]\mathbb{E}\left[\log(f_\theta(Y^{(1)}))\right] = X\mathbb{E}\left[\log(f_\theta(Y^{(1)}))\right]$$

Then, the empirical risk satisfies the following almost sure convergence :

$$R^{n2n}_{N,1}(\theta) \xrightarrow[N\to\infty]{a.s.} \mathbb{E}\left[f_\theta(Y^{(1)}) - X\log(f_\theta(Y^{(1)})) + \log(Y^{(2)}!)\right]$$

We can do the same for the second term of the loss by swapping $Y^{(1)}$ and $Y^{(2)}$ and we get the following result :

$$R^{n2n}_{N}(\theta) \xrightarrow[N\to\infty]{a.s.} 2\mathbb{E}\left[f_\theta(Y) - X\log(f_\theta(Y)) + \log(Y!)\right]$$

where we are left with the standard clean poisson negative-log likelihood risk plus a constant term independent of θ. □

In summary, having independant realizations is crucial for this framework to work. In the case of additive noise trained under the L2 loss, the noise must also be zero-mean. Either way, the method is very flexible and can be applied to various noise models. [Lehtinen et al. 2018] also mention Bernoulli noise and text removal as possible applications of their method. The previous propositions give theoretical guarantees that the estimator produced by the Noise2Noisemethod will be very close to the one produced by the supervised method, with similar variance and bias.

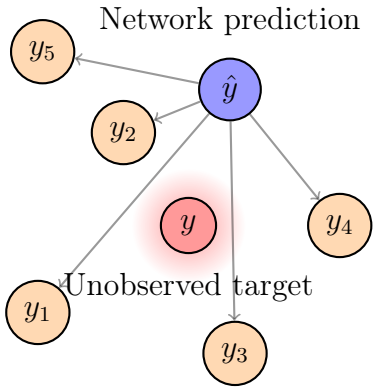

Figure 1: Illustration of the Noise2Noise framework where a neural network is trained to map a noisy input image to multiple noisy target images representing different realizations of the same underlying clean image.

2.2.3 Consensus loss

The two independant noisy observations may be obtained by splitting the data into two different subsets, hence the noise in each subset is higher than the noise in the original data. This is the case for example in the medical field. [Wu et al. 2019] propose a consensus loss to alleviate this issue and improve the results of the aggregated outputs

$$\hat{y}_i = \frac{1}{2}(f_\theta(y_i^{(1)}) + f_\theta(y_i^{(2)})) \tag{32}$$

In this section, we assume that ρ is the MSE loss. We start by stating that the supervised loss of the aggregated output can be written as follows :

$$\begin{aligned}\mathcal{L}_{\text{clean}}(\theta) &= \frac{1}{N}\sum_{i=1}^{N} \|\frac{1}{2}(f_\theta(y_i^{(1)}) + f_\theta(y_i^{(2)})) - x_i\|_2^2 \\ &= \frac{1}{N}\sum_{i=1}^{N} \frac{1}{2}\|f_\theta(y_i^{(1)}) - x_i\|_2^2 + \frac{1}{2}\|f_\theta(y_i^{(2)}) - x_i\|_2^2 - \frac{1}{4}\|f_\theta(y_i^{(1)}) - f_\theta(y_i^{(2)})\|_2^2\end{aligned}$$

Theorem 2.3 show that the the unobserved clean image x_i can be replaced by one of its noisy version $y_i^{(j)}$ in the previous expression without changing the optimal solution of the problem. Thus, we can rewrite the supervised loss of the aggregated output as follows :

$$\mathcal{L}_{\text{consensus}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\|f_\theta(y_i^{(1)}) - y_i^{(1)}\|_2^2 + \frac{1}{2}\|f_\theta(y_i^{(2)}) - y_i^{(2)}\|_2^2 - \frac{1}{4}\|f_\theta(y_i^{(1)}) - f_\theta(y_i^{(2)})\|_2^2 \tag{33}$$

Noise2NoiseFlow [Maleky et al. 2022] propose a framework at the crossroad between density estimation and the Noise2Noise framework. Density estimation is performed using Noise Flow [Abdelhamed et al. 2019], a normalizing flow model designed to learn complex noise distributions from data. Notice that the density is closely related to the score. Whereas classical score matching methods require model trained in a supervised manner with clean data or denoising auto-encoders trained in an unsupervised manner to estimate the score function, they propose to directly learn the score function from noisy data only using the Noise2Noise principle. Both the denoising model and the normalizing flow are trained jointly using pairs of noisy images under the following loss function :

$$\mathcal{L}(\theta, \phi) = \mathop{\mathbb{E}}_{y_1, y_2 \sim p_{noisy}} \left[-\log p_{N_\phi}(y_1 | f_\theta(y_2)) - \log p_{N_\phi}(y_2 | f_\theta(y_1)) \right] + \lambda \mathcal{L}_{n2n}(\theta) \tag{34}$$

with p_{N_ϕ} being the noise distribution modeled by the normalizing flow with parameters ϕ, f_θ the denoising model with parameters θ and $\mathcal{L}_{n2n}(\theta)$ the standard Noise2Noise loss. The authors of Noise2NoiseFlow show that this additional term gives more stability to the training process and leads to better denoising results.

Note that this framework is useless if we already know the natural noise distribution in the data as we could directly use it to denoise images using bayesian methods.

3 Linear inverse problems

The main focus of this section is to present mathematical inverse problems, which are problems where we aim to recover an underlying signal from indirect and noisy measurements. These problems are common in various fields such as image reconstruction, signal processing, and machine learning. We limit ourselves to linear inverse problems in finite dimensions. This allows us to derive theoretical results and develop efficient algorithms for solving these problems.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X : \Omega \to \mathbb{R}^n$ a random variable representing the true signal. We assume that we have access to measurements $Y : \Omega \to \mathbb{R}^d$ that are related to the true signal through a linear operator $A : \mathbb{R}^n \to \mathbb{R}^d$ and corrupted by a noise process $\mathcal{C}$. The measurement model can be written as:

$$Y|X = x \sim \mathcal{C}(Ax) \tag{35}$$

In most setups, we work in high dimensions and the forward operator A is not directly invertible, which makes the problem of recovering X from Y an ill-posed inverse problem.

Definition 3.1. *A is said to be ill-posed if one of the following properties is not satisfied :*

1. *A is injective, i.e. $\forall x_1, x_2 \in \mathbb{R}^n, Ax_1 = Ax_2 \implies x_1 = x_2$.*
2. *A is surjective, i.e. $\forall y \in \mathbb{R}^d, \exists x \in \mathbb{R}^n$ such as $y = Ax$.*
3. *A^{-1} is continuous, i.e. $\forall \epsilon > 0, \exists \delta > 0$ such as $\|y_1 - y_2\| < \delta \implies \|A^{-1}y_1 - A^{-1}y_2\| < \epsilon$.*

Remark 3.1. *Note that in finite dimention, linear bijections are always continuous and thus the third condition is automatically satisfied.*

If A is ill-posed, then it is obviously not a bijection and its null-space is not reduced to $\{0\}$. This means that there exist many solutions to the inverse problem for a given measurement, eventhough we're looking for a unique solution. In order to solve this issue, one may consider regularizing the problem by adding prior knowledge on the solution. The goal of this prior is to eliminate the spurious solutions and retrieve a signal as close as possible to the true signal.

3.1 Generalization of Noise2Noise to linear inverse problems

[Xia et al.] extend the Noise2Noise framework (see section 2.2) to linear inverse problems such as image restoration. Formalism from section 2.2 is still valid and

we will use the same notations in the following. Let again $\{(Y_i^{(1)}, Y_i^{(2)})\}_{i=1}^N$ be a finite sequence of independant and statistically distributed random variables and $y_i^{(j)} = Y_i^{(j)}(\omega_i) = A_i^{(j)} x_i + b_i^{(j)}$ the observed degraded images for $\omega \in \Omega$, $j = 1, 2$, $i = 1, \dots, N$. We also rewrite 35 as :

$$\begin{cases} Y^{(j)}|X = x \sim \mathcal{C}(A^{(j)}x), \quad j = 1, 2 \\ Y^{(j)} = A^{(j)}X + B^{(j)}, \quad j = 1, 2 \end{cases} \tag{36}$$

with $A^{(j)} \in \mathbb{R}^{d \times m}$ two linear operators inheriting from A and defined accordingly to $Y^{(1)}$ and $Y^{(2)}$.

In the context of non-blind estimation, i.e. when $A^{(j)}$ is known, we can optimize a neural network f_θ under the following objective function :

$$\mathcal{L}_{swap}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left[\rho(A_i^{(2)} f_\theta(y_i^{(1)}) - y_i^{(2)}) + \rho(A_i^{(1)} f_\theta(y_i^{(2)}) - y_i^{(1)}) \right] \tag{37}$$

The authors add a second term to the loss function to enforce data consistency with respect to the measurements y.

$$\mathcal{L}_{self} = \frac{1}{N} \sum_{i=1}^{N} \left[\rho(A_i^{(1)} f_\theta(y_i^{(1)}) - y_i^{(1)}) + \rho(A_i^{(2)} f_\theta(y_i^{(2)}) - y_i^{(2)}) \right]$$

The final loss function is then written as :

$$\mathcal{L}_{total} = \mathcal{L}_{swap} + \lambda \mathcal{L}_{self} \tag{38}$$

with λ a hyperparameter to tune the weight of the data-consistency term.

Remark 3.2. *Note that choosing $I = A_i^{(1)} = A_i^{(2)} \, \forall i$ is equivalent to the regular Noise2Noise framework.*

Theorem 3.1. *Minimizing 30 with $\rho(x, y) = \|x - y\|_2^2$ gives the same estimator as minimizing the homologous supervised loss as $N \to \infty$ under Noise2Noise statistical conditions.*

Proof. Proof in theorem 2.3 is still valid as long as we replace $f_\theta(Y^{(j)})$ by $Af_\theta(Y^{(j)})$. We thus have :

$$\begin{aligned} R_N^{n2n}(\theta) &= \frac{1}{N} \sum_{i=1}^{N} \left[\|A_i^{(2)} f_\theta(Y_i^{(1)}) - Y_i^{(2)}\|_2^2 + \|A_i^{(1)} f_\theta(Y_i^{(2)}) - Y_i^{(1)}\|_2^2 \right] \\ &= 2R_N^{clean}(\theta) + \frac{1}{N} \sum_{i=1}^{N} \left[\|B_i^{(1)}\|_2^2 + \|B_i^{(2)}\|_2^2 \right] + o(1) \end{aligned}$$

□

3.1.1 Poisson linear inverse problems

With the conventions of section 2.2.2, we now consider the case of linear inverse problems corrupted by Poisson noise :

$$Y^{(j)} \sim Poisson(A^{(j)}X), \quad j = 1, 2 \tag{39}$$

Theorem 3.2. *If $Y|X = x \sim Poisson(Ax)$, then minimizing 30 with $\rho = \mathcal{L}_{Poisson}$ gives the same estimator as minimizing the homologous supervised loss as $N \to \infty$ under Noise2Noise statistical conditions.*

Proof. Proof in theorem 2.4 is still valid as long as we replace $f_\theta(Y^{(j)})$ by $Af_\theta(Y^{(j)})$. If we define the empirical risk as :

$$R_N^{n2n}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\mathcal{L}_{Poisson}(A_i^{(2)}f_\theta(Y_i^{(1)}), Y_i^{(2)}) + \mathcal{L}_{Poisson}(A_i^{(1)}f_\theta(Y_i^{(2)}), Y_i^{(1)})\right]$$

then we have :

$$R_N^{n2n}(\theta) \xrightarrow[N\to\infty]{a.s.} 2\mathbb{E}\left[f_\theta(Y) - AX\log(f_\theta(Y)) + \log(Y!)\right]$$

□

References

Abdelhamed, Abdelrahman et al. (Aug. 2019). *Noise Flow: Noise Modeling with Conditional Normalizing Flows*. arXiv:1908.08453 [cs]. DOI: 10.48550/arXiv.1908.08453. URL: http://arxiv.org/abs/1908.08453 (visited on 12/05/2025).

Alain, Guillaume et al. (Aug. 2014). *What Regularized Auto-Encoders Learn from the Data Generating Distribution*. arXiv:1211.4246 [cs]. DOI: 10.48550/arXiv.1211.4246. URL: http://arxiv.org/abs/1211.4246 (visited on 11/03/2025).

Batson, Joshua et al. (n.d.). "Noise2Self: Blind Denoising by Self-Supervision". en. In: ().

Efron, Bradley (Dec. 2011). "Tweedie's Formula and Selection Bias". en. In: *Journal of the American Statistical Association* 106.496, pp. 1602–1614. ISSN: 0162-1459, 1537-274X. DOI: 10.1198/jasa.2011.tm11181. URL: http://www.tandfonline.com/doi/abs/10.1198/jasa.2011.tm11181 (visited on 10/28/2025).

Hudson, H.M. et al. (1994). "Accelerated image reconstruction using ordered subsets of projection data". In: *IEEE Transactions on Medical Imaging* 13.4, pp. 601–609. ISSN: 0278-0062. DOI: 10.1109/42.363108. URL: http://dx.doi.org/10.1109/42.363108.

Hyvärinen, Aapo (2005). "Estimation of Non-Normalized Statistical Models by Score Matching". In: *Journal of Machine Learning Research* 6.24, pp. 695–709. URL: http://jmlr.org/papers/v6/hyvarinen05a.html.

Kim, Kwanyoung et al. (n.d.). "Noise2Score: Tweedie's Approach to Self-Supervised Image Denoising without Clean Images". en. In: ().

Krull, Alexander et al. (Apr. 2019). *Noise2Void - Learning Denoising from Single Noisy Images*. arXiv:1811.10980 [cs]. DOI: 10.48550/arXiv.1811.10980. URL: http://arxiv.org/abs/1811.10980 (visited on 11/03/2025).

Lehtinen, Jaakko et al. (Oct. 2018). *Noise2Noise: Learning Image Restoration without Clean Data*. en. arXiv:1803.04189 [cs]. DOI: 10.48550/arXiv.1803.04189. URL: http://arxiv.org/abs/1803.04189 (visited on 10/10/2025).

Maleky, Ali et al. (June 2022). *Noise2NoiseFlow: Realistic Camera Noise Modeling without Clean Images*. arXiv:2206.01103 [eess]. DOI: 10.48550/arXiv.2206.01103. URL: http://arxiv.org/abs/2206.01103 (visited on 12/05/2025).

Moran, Nick et al. (Oct. 2019). *Noisier2Noise: Learning to Denoise from Unpaired Noisy Data*. arXiv:1910.11908 [eess]. DOI: 10.48550/arXiv.1910.11908. URL: http://arxiv.org/abs/1910.11908 (visited on 11/03/2025).

Nuyts, J. et al. (Feb. 2002). "A concave prior penalizing relative differences for maximum-a-posteriori reconstruction in emission tomography". In: *IEEE Transactions on Nuclear Science* 49.1, pp. 56–60. ISSN: 1558-1578. DOI: 10.1109/tns.2002.998681. URL: http://dx.doi.org/10.1109/TNS.2002.998681.

Rudin, L.I. et al. (1992). "Total variation based image restoration with free local constraints". In: *Proceedings of 1st International Conference on Image Processing*. Vol. 1. ICIP-94. IEEE Comput. Soc. Press, pp. 31–35. DOI: 10.1109/icip.1994.413269. URL: http://dx.doi.org/10.1109/ICIP.1994.413269.

Shepp, L. A. et al. (Oct. 1982). "Maximum Likelihood Reconstruction for Emission Tomography". In: *IEEE Transactions on Medical Imaging* 1.2, pp. 113–122. ISSN: 1558-254X. DOI: 10.1109/tmi.1982.4307558. URL: http://dx.doi.org/10.1109/TMI.1982.4307558.

Tikhonov, Andrey N. et al. (1977). *Solutions of ill-posed problems*. Translated from the Russian, Preface by translation editor Fritz John, Scripta Series in Mathematics. Washington, D.C.: John Wiley & Sons, New York: V. H. Winston & Sons, pp. xiii+258.

Vincent, Pascal (July 2011). "A Connection Between Score Matching and Denoising Autoencoders". en. In: *Neural Computation* 23.7, pp. 1661–1674. ISSN: 0899-7667, 1530-888X. DOI: 10.1162/NECO_a_00142. URL: https://direct.mit.edu/neco/article/23/7/1661-1674/7677 (visited on 10/28/2025).

Vincent, Pascal et al. (2008). "Extracting and composing robust features with denoising autoencoders". en. In: *Proceedings of the 25th international conference on Machine learning - ICML '08*. Helsinki, Finland: ACM Press, pp. 1096–1103. ISBN: 978-1-60558-205-4. DOI: 10.1145/1390156.1390294. URL: http://portal.acm.org/citation.cfm?doid=1390156.1390294 (visited on 11/07/2025).

Wu, Dufan et al. (2019). *Consensus Neural Network for Medical Imaging Denoising with Only Noisy Training Samples*. DOI: 10.48550/ARXIV.1906.03639. URL: https://arxiv.org/abs/1906.03639.

Xia, Zhihao et al. (n.d.). "Training Image Estimators without Image Ground Truth". en. In: ().